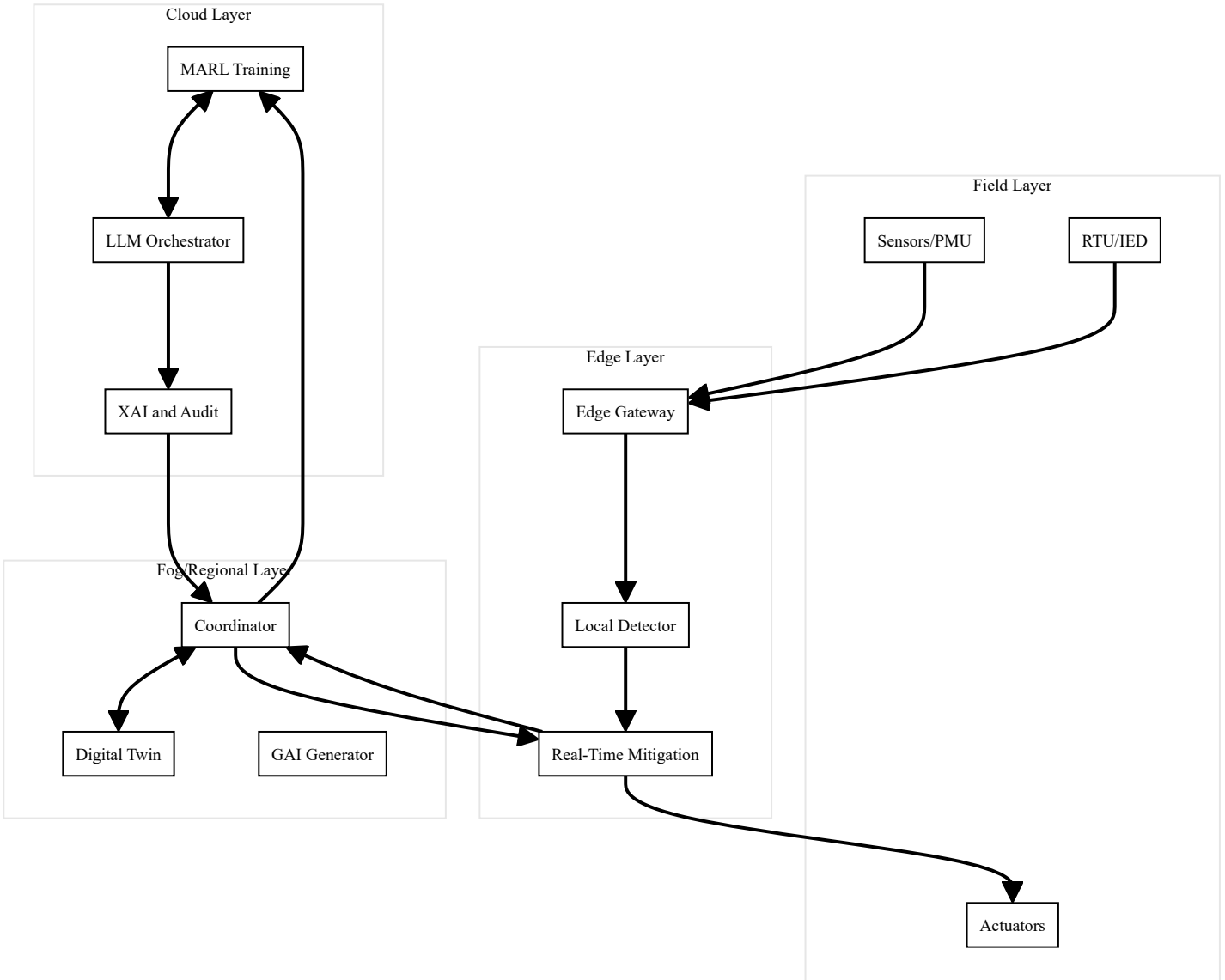
Cloud Layer
MARL Training
LLM Orchestrator
XAI and Audit
Fog/Regional Layer
Coordinator
Digital Twin
GAI Generator
Edge Layer
Edge Gateway
Local Detector
Real-Time Mitigation
Field Layer
Sensors/PMU
RTU/IED
Actuators